



MORINGA SCHOOL · DATA SCIENCE CAPSTONE · JUNE–JULY 2026

Predicting Acute Food Insecurity Risk in Kenyan Counties

Using climate and market signals as leading indicators of IPC Phase 3 (Crisis), an early-warning model for Kenya's National Drought Management Authority

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Kenya's food security response is reactive



10M+

Kenyans live in arid & semi-arid lands (ASAL), dependent on two seasonal rains



5

consecutive failed rainy seasons (2020–23), the worst drought in 40 years



4.5M

Kenyans pushed into acute food insecurity at the drought's peak



4–8 wks

lag before official IPC classifications confirm a crisis on the ground

By the time a county is officially classified in Crisis, households have already skipped meals, sold productive assets, and exhausted their coping capacity.

Who this serves, and why it matters



Kenya NDMA

County-level crisis probabilities weeks before IPC release, enabling early warnings and pre-positioned food aid.



WFP Kenya

Longer planning horizon for procurement, logistics coordination, and partner briefings.



Humanitarian NGOs

Target pre-crisis interventions to high-risk zones before deterioration passes the point of prevention.



Ministry of Agriculture

Prioritise drought-resistant seed and emergency livestock support to flagged counties ahead of a season.

Impact: a shift from reactive to predictive response: earlier mobilisation, precise county-level targeting, cheaper prevention over emergency response, and an architecture scalable across East Africa.

Crises are confirmed after the damage, yet two signals arrive weeks earlier



Today: detection, not prediction

- IPC Phase 3 (Crisis) is the official threshold at which NDMA declares a food emergency and activates aid.
- Classifications rely on field assessments and are published 4–8 weeks after conditions have deteriorated.
- Aid agencies get no lead time to pre-position resources, so response arrives after coping capacity is gone.

Two leading signals, currently unused in any predictive model



Rainfall failure: CHIRPS satellite anomalies every 10 days, weeks before crop and livestock losses are visible.



Food price spikes: WFP monthly market prices reveal collapsing purchasing power before hunger becomes measurable.

Prediction question: Can rainfall anomalies + food prices predict whether a Kenyan county reaches IPC Phase 3+ in a given month, before the official classification is published?

OBJECTIVES

One primary objective, four that make it real



Prove the signals

Demonstrate that satellite rainfall anomalies and regional food prices together carry enough predictive power to flag impending crises at the county level.



Optimize for recall

Build a recall-optimized classifier that minimizes missed crises, since a false negative carries far greater humanitarian cost than a false positive.



Explain the drivers

Use SHAP analysis to identify which climate and market signals drive food insecurity in arid and semi-arid counties, interpretable directly by NDMA analysts.



Make it reproducible

Establish an open-access pipeline that reruns monthly as new CHIRPS and WFP data are published, supporting ongoing early-warning operations.

Primary: predict acute food insecurity risk in Kenyan counties before official IPC classifications are published, enabling NDMA to issue early warnings and pre-position food aid.

Three open datasets, merged to a county-month grain



TARGET

27,694 rows

FEWS NET IPC Classifications

Zone-level food insecurity severity, Phase 1 (Minimal) to 5 (Famine). Our target: Phase 3+ = crisis.



CLIMATE FEATURES

132,678 rows

CHIRPS Rainfall Anomalies

Dekadal (10-day) satellite rainfall incl. rfq, rainfall as % of long-term average, the key drought signal.



MARKET FEATURES

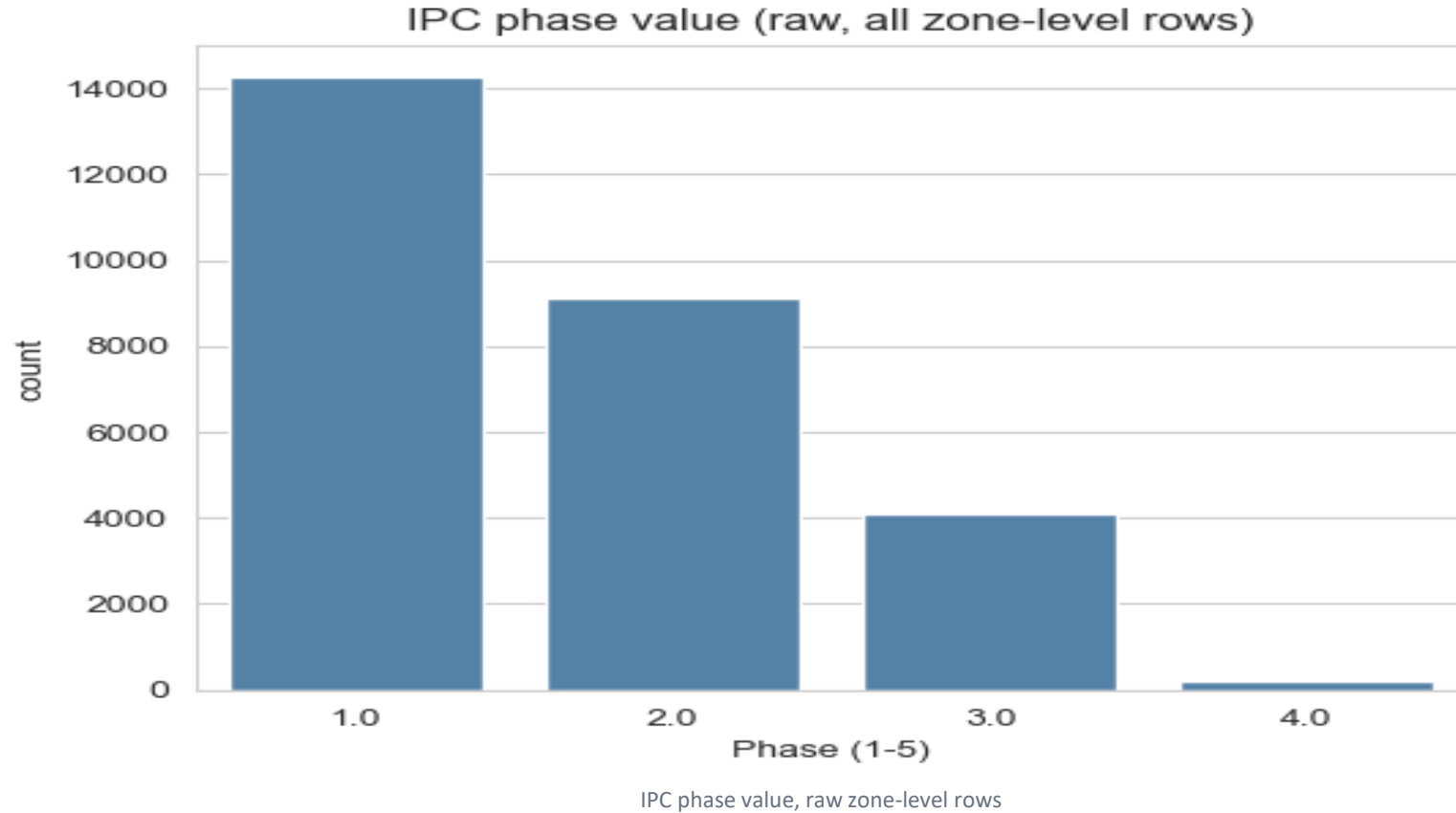
26,745 rows

WFP Food Prices, Kenya

Monthly market prices for staple commodities (maize dominant) across Kenyan markets.

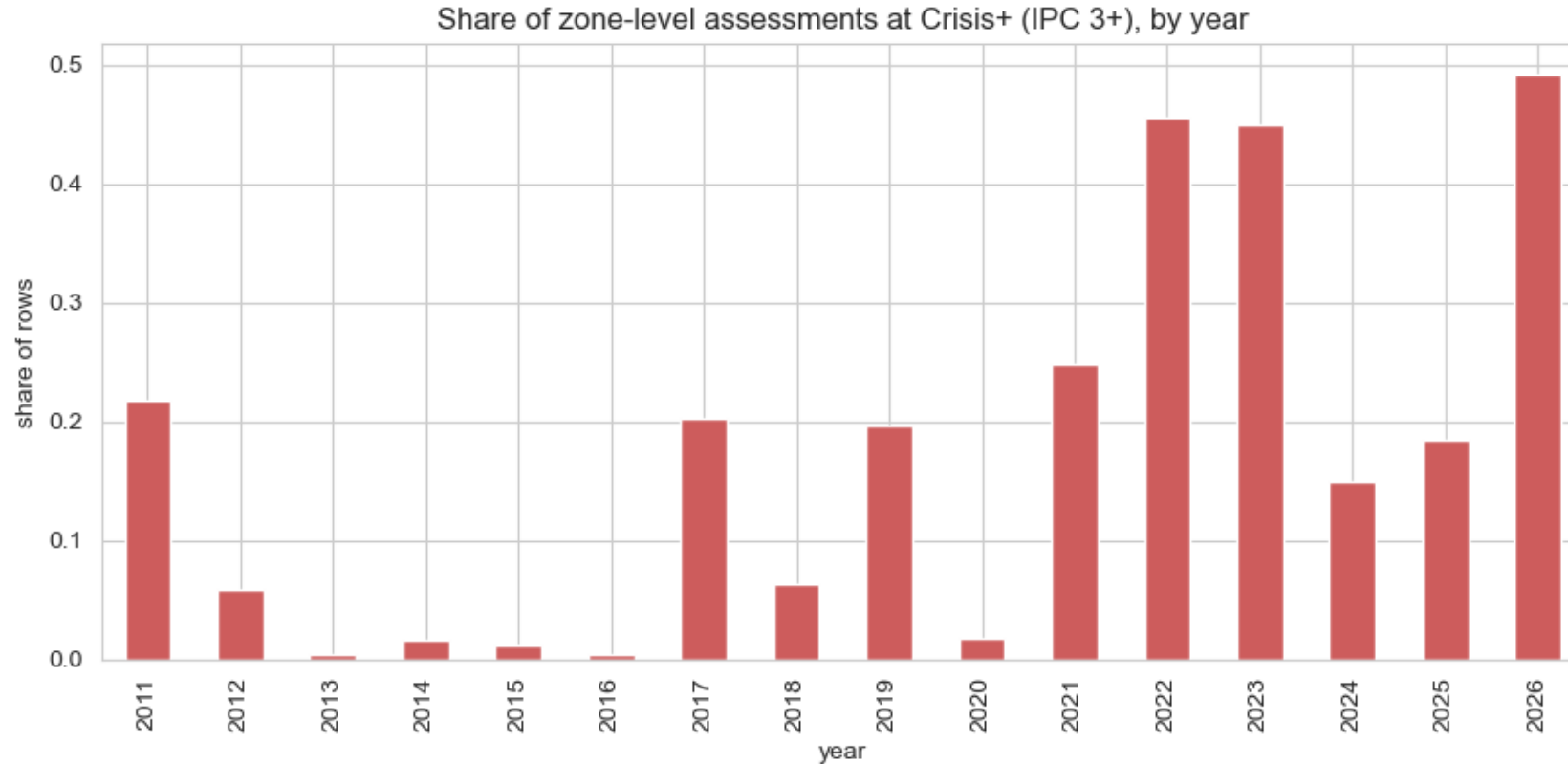
Preparation highlights: geographic harmonisation to counties · binary target construction (IPC 3+) · aggregation to county-month · lagged merge of features so predictors precede the outcome (leakage fix).

The target skews heavily toward "no crisis"



Insight: IPC phases skew heavily to Phase 1 (Minimal); Phase 3–4 are rare and Phase 5 is absent. This confirms the target is naturally imbalanced, so evaluation must lead with recall rather than accuracy.

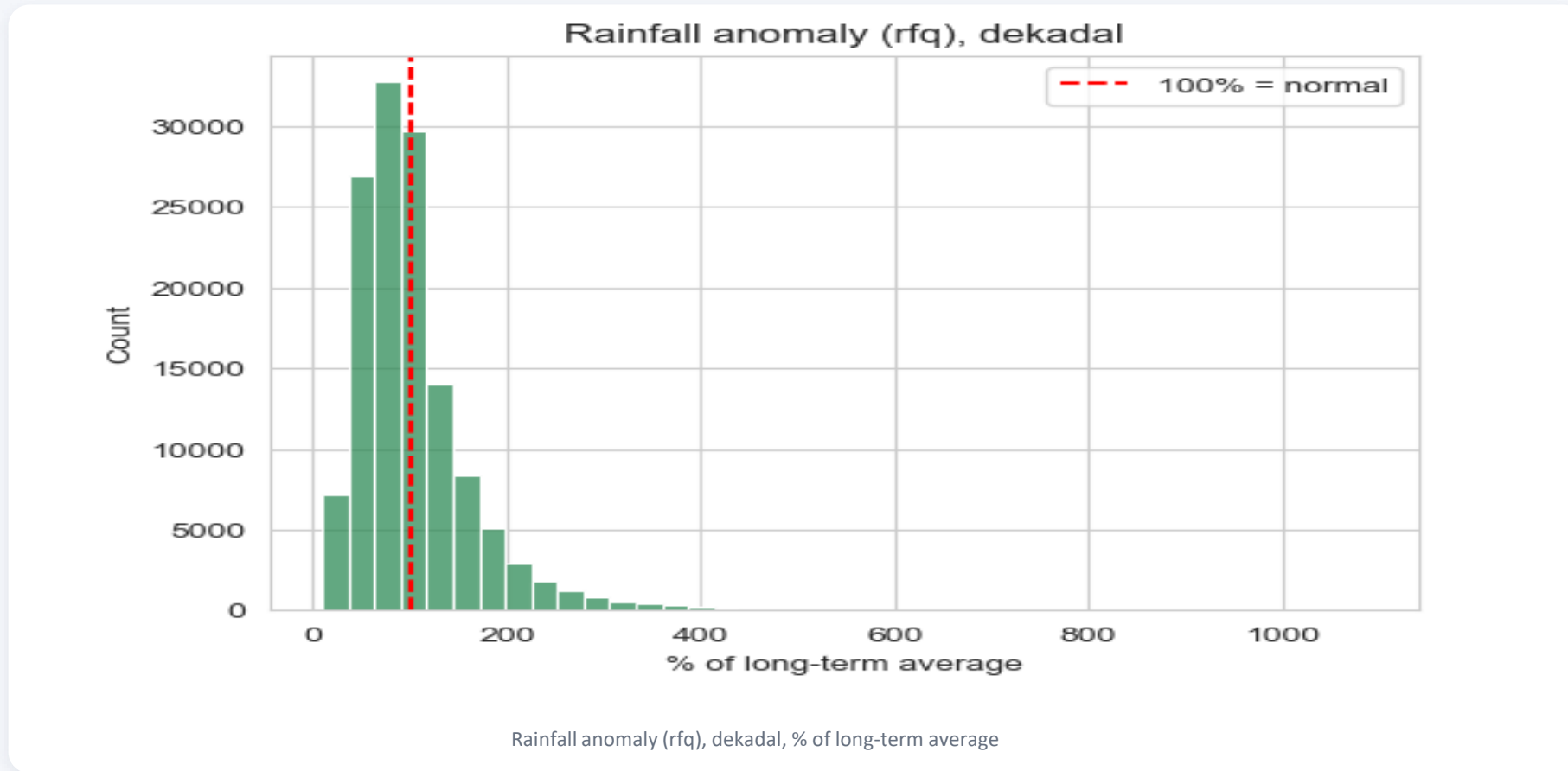
Crisis rates spike in Kenya's known drought years



Share of zone-level assessments at Crisis+ (IPC 3+), by year

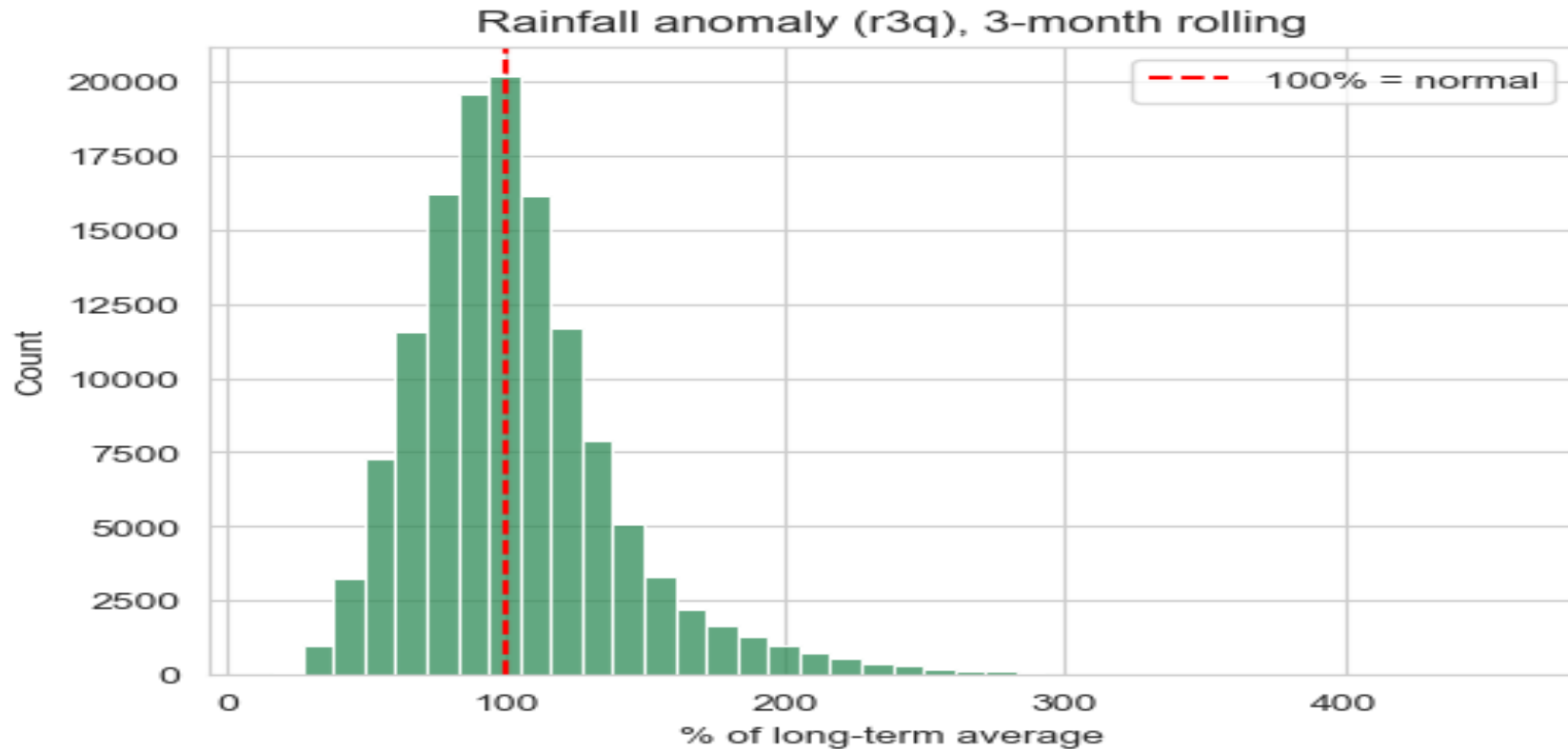
Insight: Crisis rates surge sharply in 2011, 2017 and 2021–22, Kenya's documented drought years. This confirms the target tracks real climate shocks rather than noise, making it a reliable signal to model.

Rainfall is mostly normal, with occasional extremes



Insight: The dataset is dominated by normal rainfall conditions but still includes drought and unusually wet periods. This matters because the model needs examples of both normal and extreme conditions to learn how rainfall influences food insecurity.

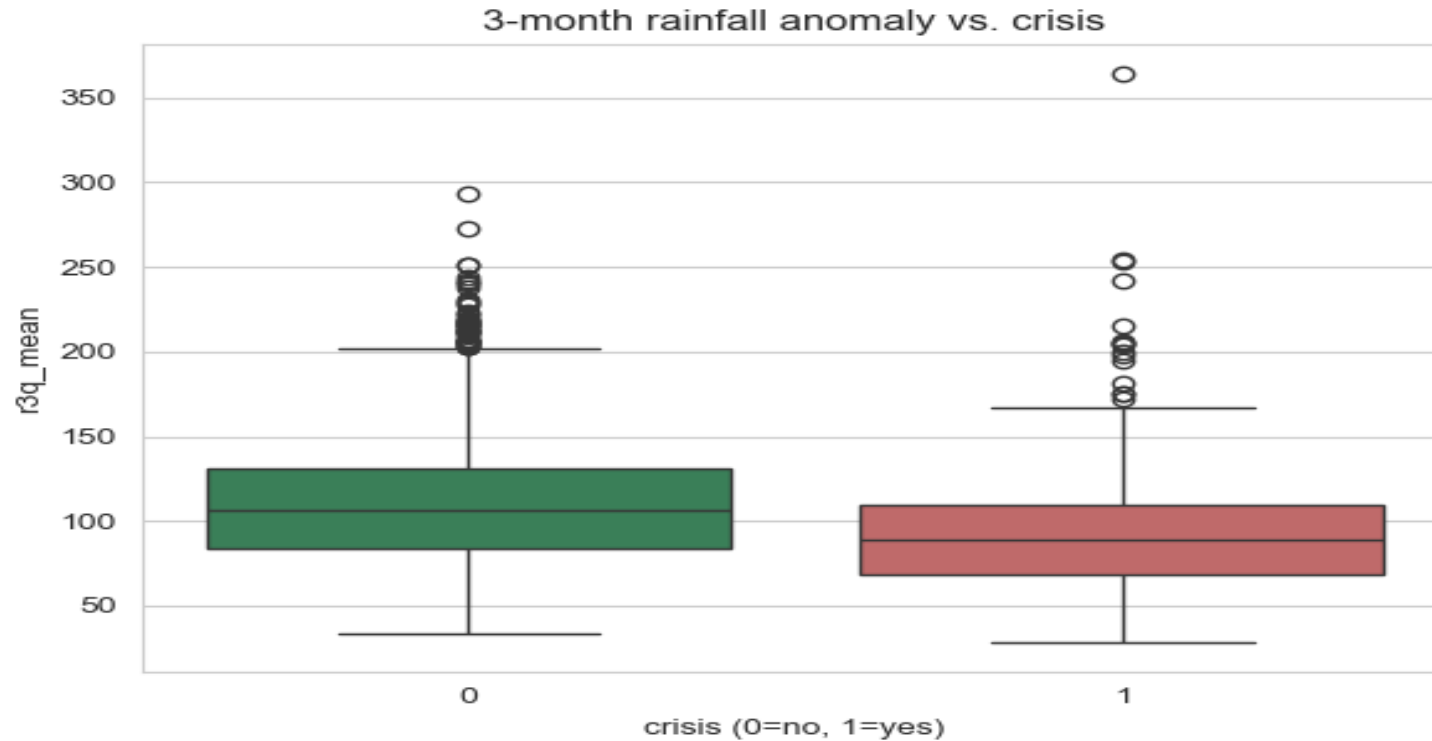
The 3-month window gives a steadier drought signal



Rainfall anomaly (r3q), 3-month rolling, % of long-term average

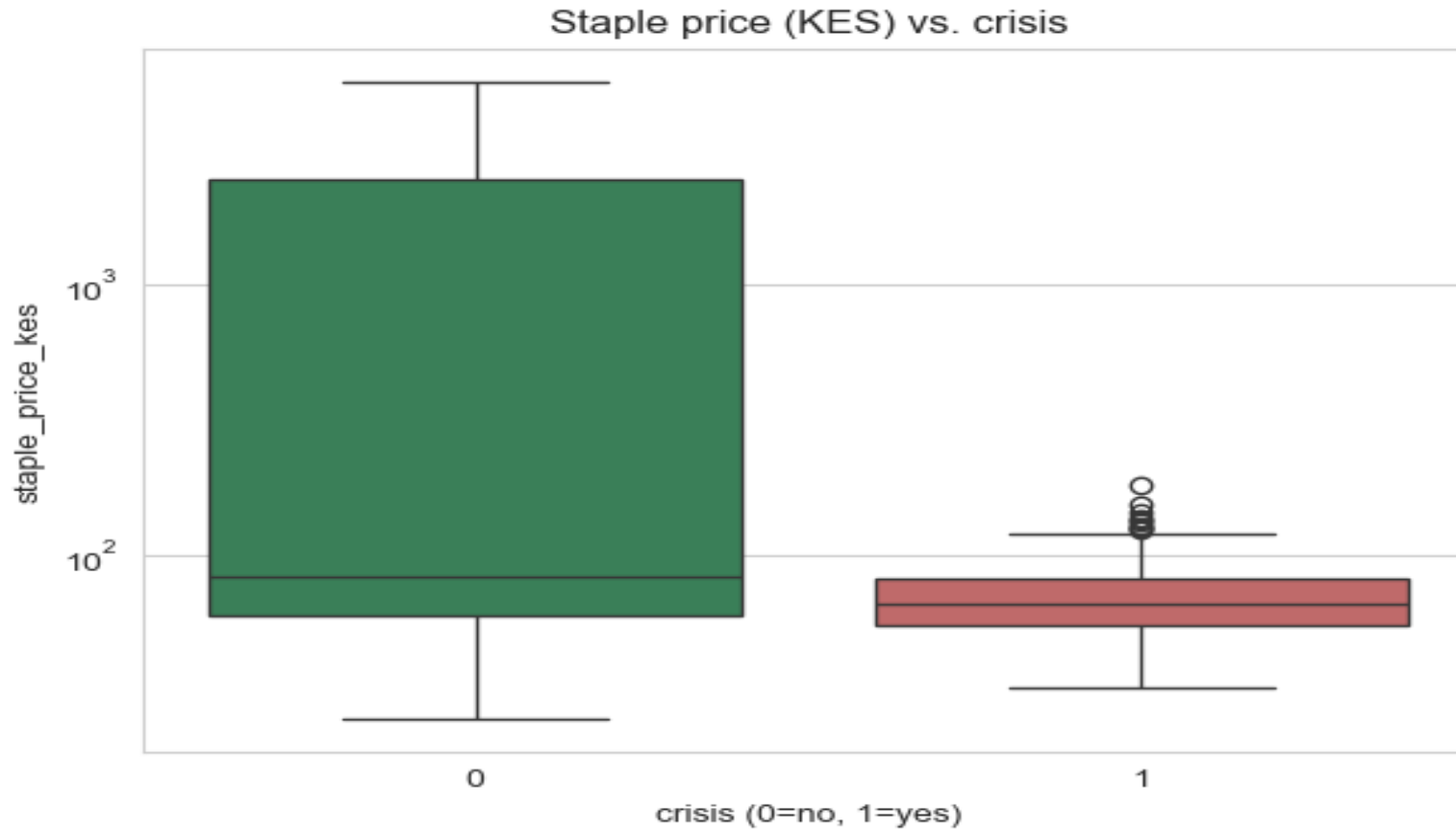
Insight: r3q summarizes rainfall over the previous three months rather than a single 10-day period, giving a more stable indicator of agricultural stress. Most observations stay close to the 100% average, with fewer observations representing prolonged dry or unusually wet conditions.

Crisis counties show clearly lower rainfall anomalies



Insight: Counties in crisis generally show lower rainfall anomalies than non-crisis periods, supporting rainfall deficits as an early-warning signal. The 3-month window separates the two groups more clearly than a single short-term reading, since agricultural stress accumulates over time.

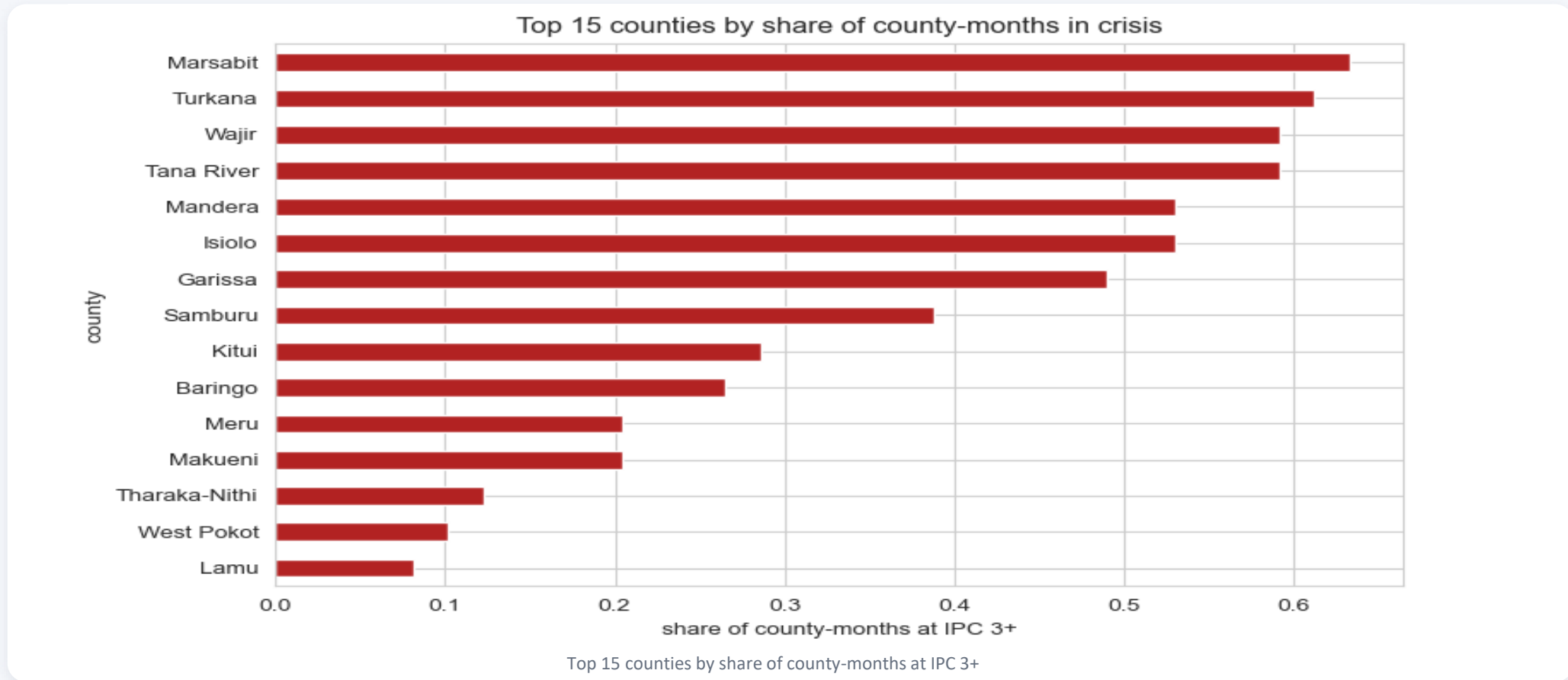
Price level alone does not separate crisis from calm



Staple price (KES), log scale, vs crisis outcome

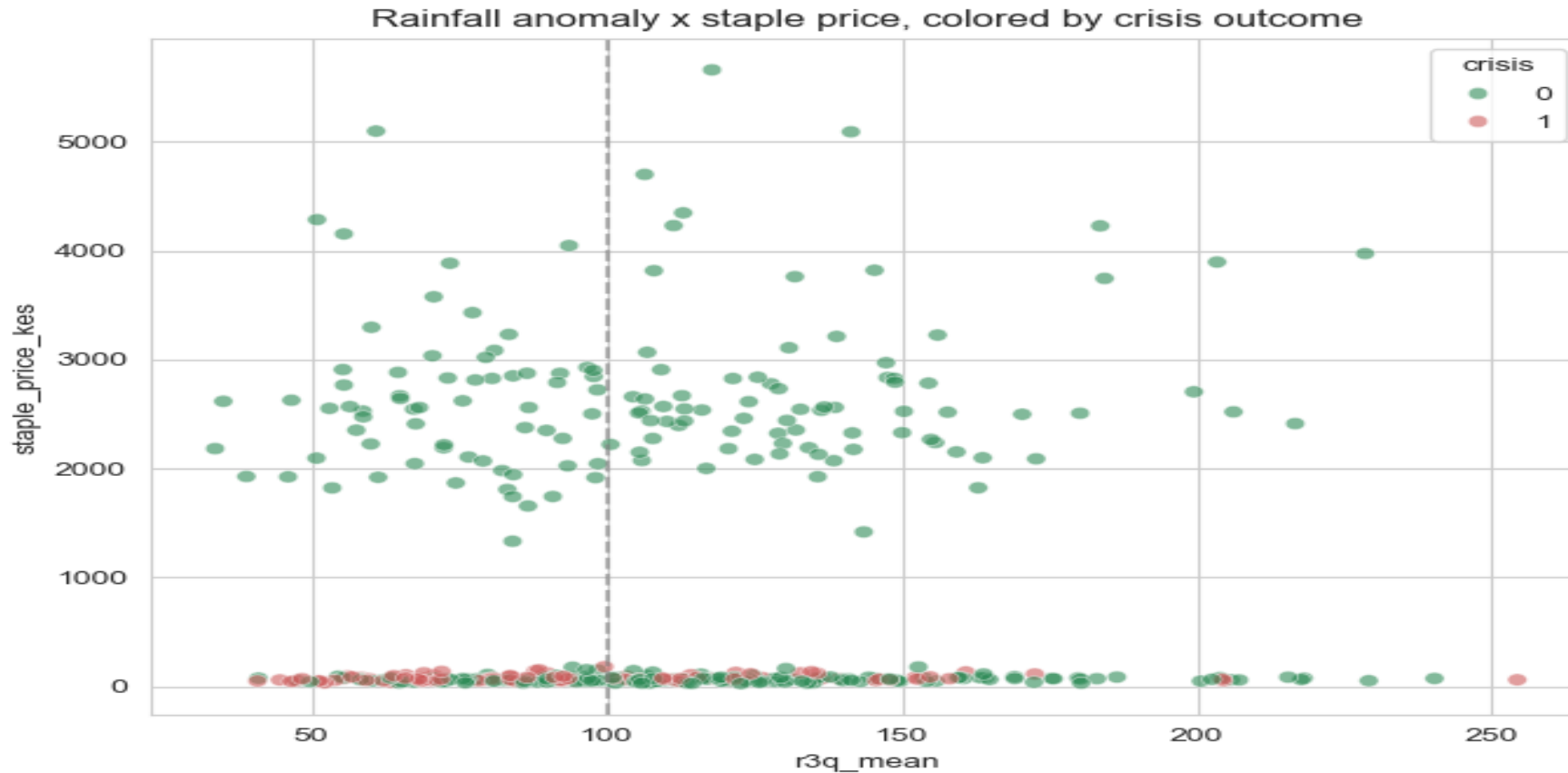
Insight: Contrary to expectation, staple prices are not consistently higher during crisis periods, the non-crisis group even shows a higher median. Price is shaped by many factors besides food insecurity, such as transport costs and supply chains, making it a supporting signal rather than a standalone predictor.

Risk concentrates in Kenya's arid and semi-arid counties



Insight: The highest-crisis counties, Turkana, Mandera, Marsabit, Wajir, Garissa and Samburu, sit in Kenya's arid and semi-arid lands. This geographic clustering reflects known vulnerability patterns and justifies including an `is_asal` feature in the model.

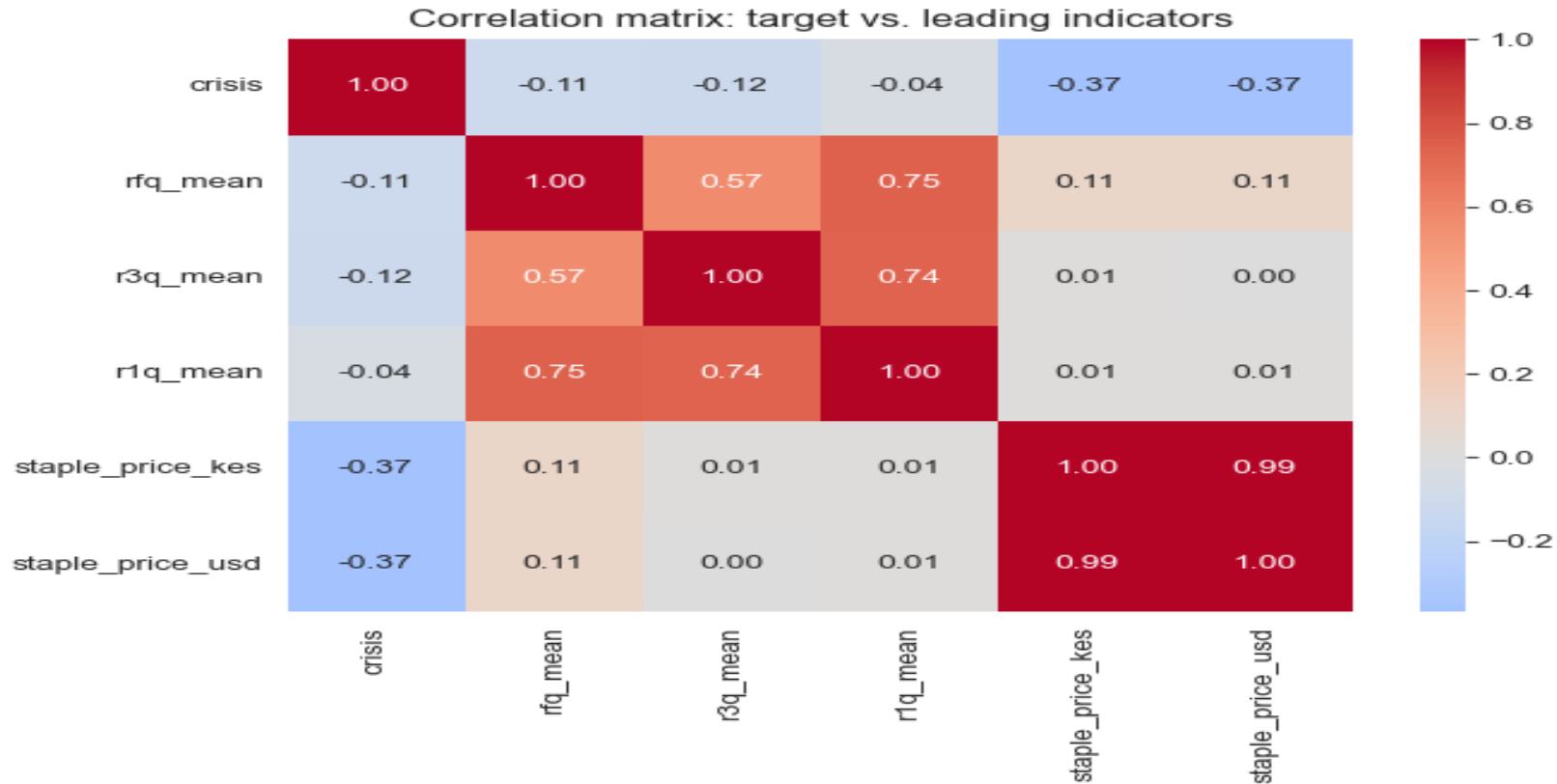
Crisis clusters where drought and high prices coincide



Rainfall anomaly (r3q) x staple price, colored by crisis outcome

Insight: Crisis observations concentrate where rainfall is below normal and prices are elevated at the same time. This shows food insecurity is usually a compound shock, reduced availability and reduced affordability together, supporting the case for including both signal families.

Rainfall correlates with crisis far more than price does



Insight: Rainfall variables show the strongest (negative) correlation with crisis; price variables are weaker. The rainfall variables also correlate with each other since they capture the same climate signal at different time windows, but each still adds useful information.

Built for a public agency, not a tech lab



Tested the way it will really be used

We trained the model on older data and tested it only once, on months it had never seen before, exactly how it will work once deployed.



We tried many approaches, kept the practical ones

We compared ten different modeling approaches and kept the four that are simple, fast, and easy to run on an ordinary computer. More complex options were dropped because they would be too costly to maintain.



One clear rule for when to raise an alarm

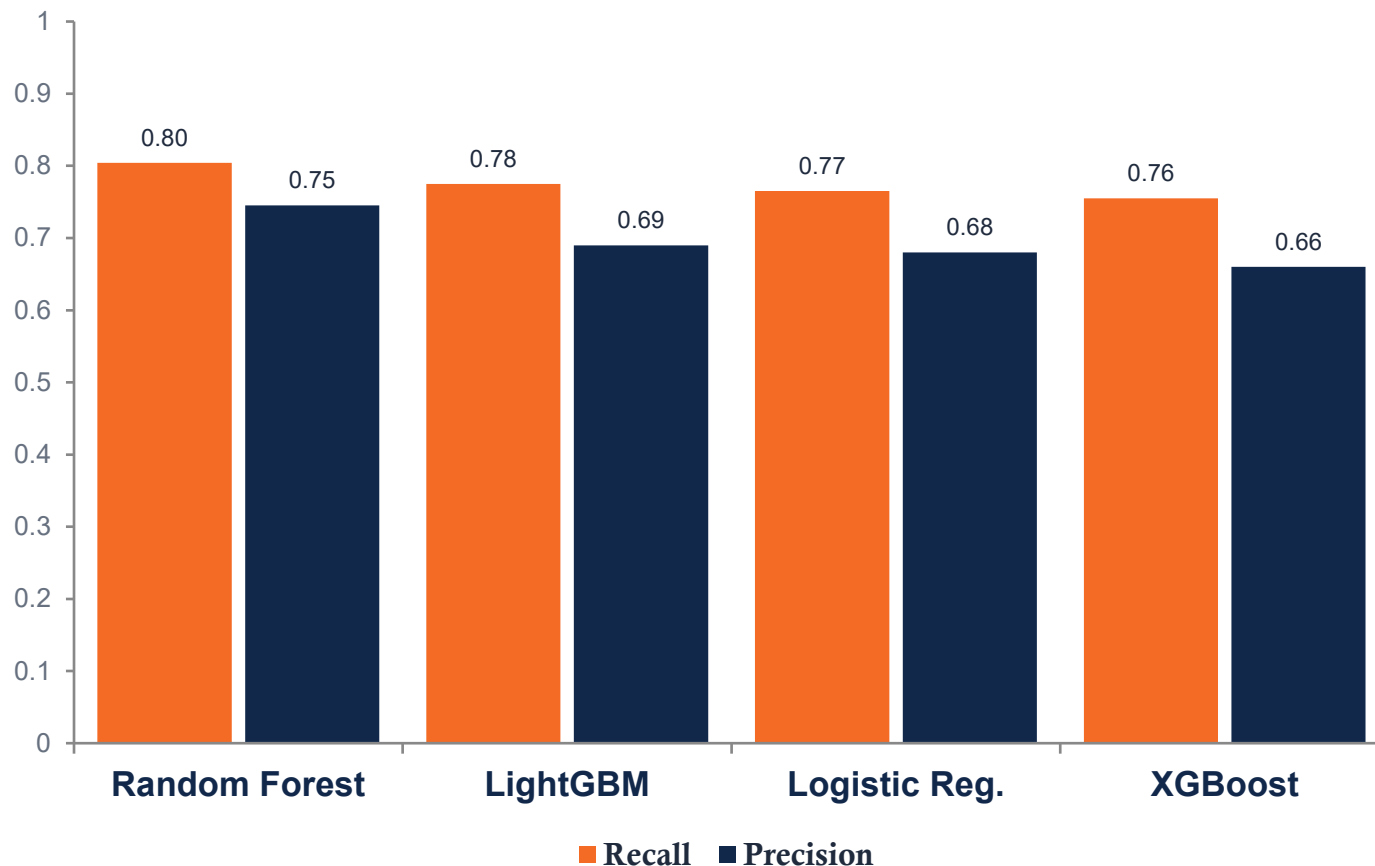
The model must always catch at least three out of every four real crises. Among the options that meet this bar, we picked the one with the fewest false alarms.

Why it matters: If the system cries wolf too often, people stop trusting it. Keeping false alarms low is what makes the early warnings useful in practice.

RESULTS

Random Forest wins, every pre-set target met on untouched test data

Test set (2022+): recall & precision by model



Random Forest, the deployed model

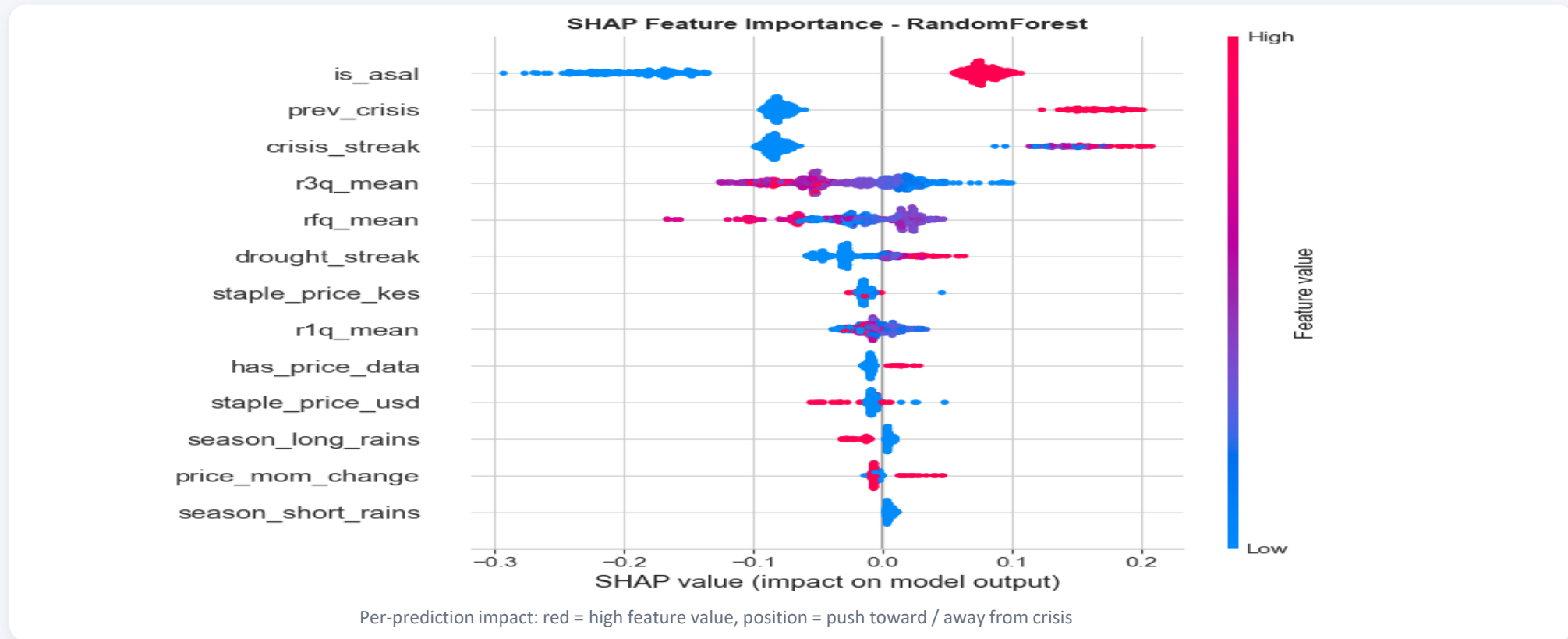
- All 6 success criteria met: Recall 0.80 · Precision 0.75 · PR-AUC 0.82 · F1-macro 0.86 · ROC-AUC 0.94, far above the 0.197 naive baseline.
- 82 of 102 crises caught with only 28 false alarms, roughly 3 crises caught per false alarm, an alert stream analysts can trust.
- Live demo, October 2023: all 8 counties that reached Crisis+ were caught and ranked at the top of the bulletin.

Targets set before any modelling began

Metric	Target	Rationale
Recall (crisis class)	≥ 0.75	Primary metric, since a missed crisis costs far more than an unnecessary alert
Precision (crisis class)	≥ 0.60	Prevents alert fatigue and wasted field-verification trips

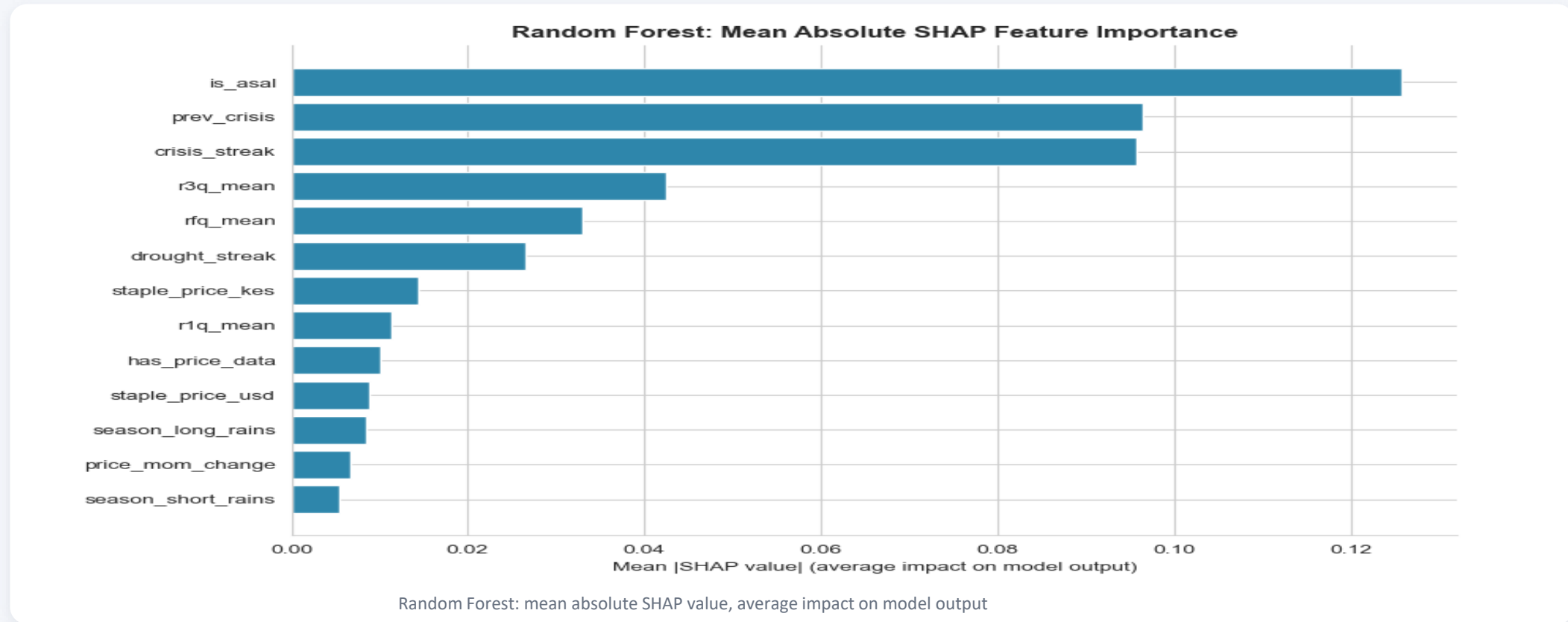
Recall first, precision second. A false negative, failing to flag a county heading into crisis, carries far greater humanitarian cost than a false alarm.

What pushes a county toward crisis (SHAP)



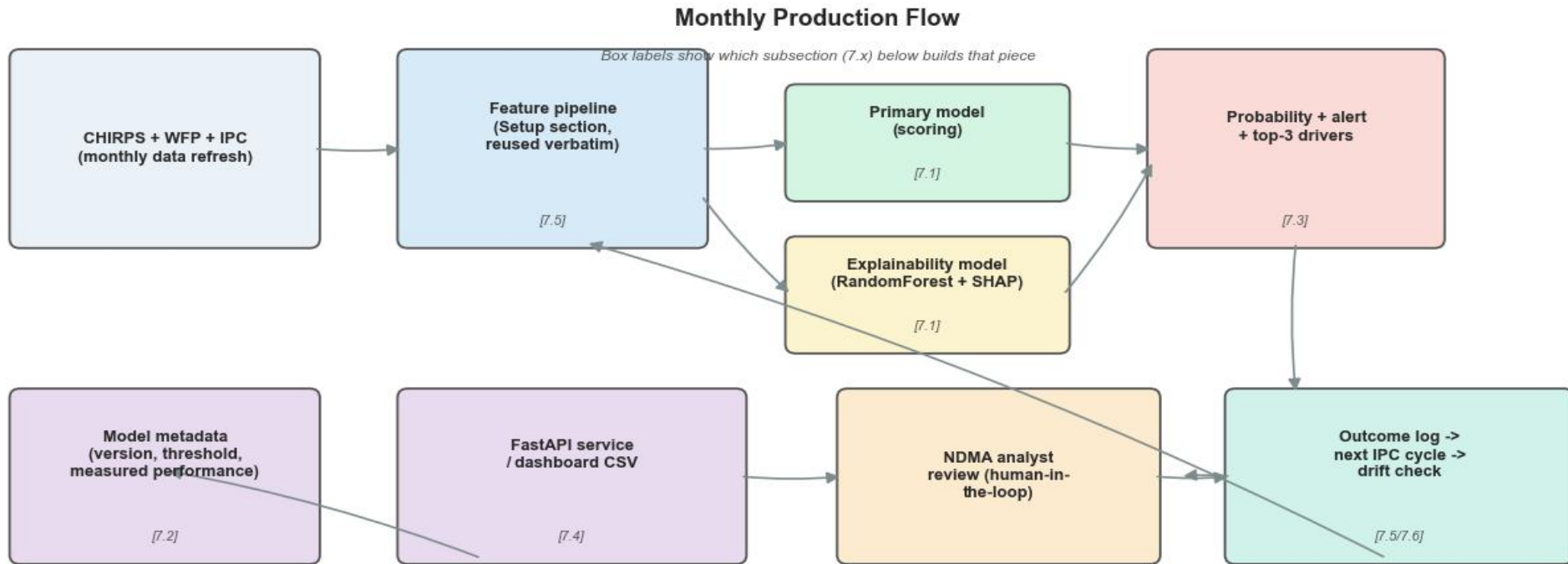
What this shows: is_asal dominates, with crisis_streak and prev_crisis next: structural vulnerability and recent history push most strongly toward crisis. Rainfall (r3q, rfq) is the strongest condition-based signal, while price level barely contributes.

Which features matter most, on average (SHAP)



What this shows: Averaged across all predictions, the ranking confirms the same story: `is_asal` leads, followed by recent crisis history and rainfall. This matches the beeswarm plot and the EDA, reinforcing that price level is not a major independent driver.

A monthly scoring pipeline with an analyst in the loop



Data flows left to right through scoring (top row), then down into serving, analyst review, and outcome logging

One run per month: new CHIRPS and WFP data refresh feeds the exact training-time feature pipeline; the model scores every county; SHAP attaches the top-3 drivers to each flag; the analyst reviews before any action is taken.

Lightweight, versioned, and monitored



Versioned artifacts

Deployed model, fitted imputer and feature order saved as small joblib files, alongside a metadata JSON recording version, decision threshold and measured test performance.



Scoring service

A FastAPI service (or a plain dashboard CSV export) exposes `score_new_month`, the exact function a scheduled job calls; runs on any laptop CPU, no GPU or MLOps stack.



Human-in-the-loop dashboard

Each county lands on the NDMA dashboard as a crisis probability, an alert flag and a plain-language top-3 driver explanation. Analysts review; the model never triggers aid directly.



Monitoring & drift checks

Every prediction is logged and scored against the official IPC release 4–8 weeks later, catching performance drift before it becomes a missed crisis; monthly retraining slots in as new data lands.

Total footprint: a few MB of serialized files plus a scheduled monthly job, a realistic recurring cost for a public agency.

Scoring a real month: the October 2023 bulletin

County	Crisis probability	Actual outcome
Wajir	94.1%	✓ Crisis
Nyeri	93.8%	No crisis
Samburu	93.2%	✓ Crisis
Turkana	92.6%	✓ Crisis
Tana River	90.2%	✓ Crisis
Marsabit	90.1%	✓ Crisis
Mandera	89.8%	✓ Crisis
Garissa	89.4%	✓ Crisis
Tharaka-Nithi	87.2%	No crisis
Isiolo	85.8%	✓ Crisis

Top 10 of 18 flagged counties (of 47 scored), ranked by predicted probability.



8 / 8

counties that truly reached Crisis+ that month were caught, all ranked at the top of the bulletin



18 / 47

counties flagged for analyst review, a manageable verification workload



10

false alarms, concentrated in counties with genuinely elevated risk factors rather than spread randomly

Analysts open a ranked bulletin, not a model file.

What the model tells NDMA



Structural vulnerability dominates

is_asal is the single strongest driver: being an arid/semi-arid county alone pushes strongly toward crisis.



Rainfall is the top condition signal

Low r3q / rfq push toward crisis, as hypothesised, and the strongest of the “what’s changing now” features.



History matters

crisis_streak and prev_crisis rank next: recent crisis history carries independent predictive signal.



Price level barely contributes

Consistent with EDA: price trend, not level, is the clearer path for future features.



Honest limitation: the top drivers describe where a county is and what it already was, not what is currently changing. The model excels at confirming known-vulnerable counties and is weaker at catching a new crisis in a previously stable one, a caveat we carry forward, not a reason to discount the result.

From notebook to NDMA operations



Deploy as a monthly early-warning bulletin

Scheduled job rebuilds features as new CHIRPS & WFP data land; each county gets a crisis probability + a plain-language SHAP explanation on an NDMA dashboard.



Keep the analyst in the loop

Flags inform; they never trigger aid directly. Log every prediction and score it against the IPC release 4–8 weeks later to catch drift before it costs a missed crisis.



Next feature work: dynamics over statics

Add price trends (not levels) and longer rolling rainfall windows to improve detection of new crises emerging in previously stable counties.



Scale the architecture

CHIRPS, WFP prices and FEWS NET IPC exist across East Africa, so the same pipeline can extend beyond Kenya.

The prediction problem is solved to the standard NDMA needs; what remains is operational integration.

Thank you.

From reactive response to predictive protection, weeks earlier, for the counties that need it most.

Questions & discussion